Asset Group

Seating: sofa, armchair, recliner

Lighting: floor lamp, table lamp, chandelier

Decor & Accessories: curtains, cushions, throws

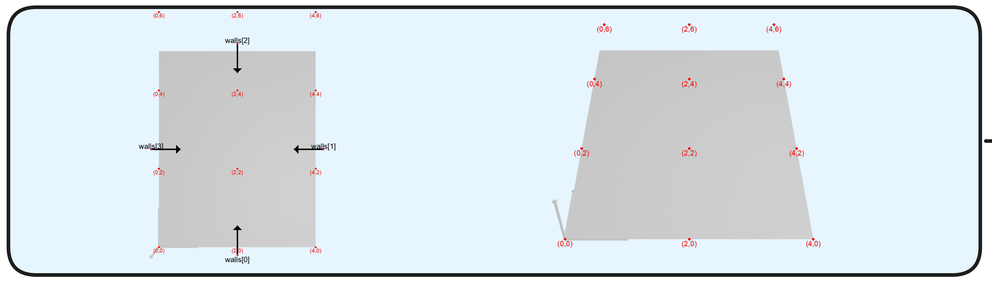
Electronics & Appliances: TV, speaker, fan

Language Instruction

Quiet and Focused Office; Place desk near the window.

Warm and Welcoming Entry with a bench and shoe storage.

Scence and Visual marks



VLM

Pose Estimate

Place a side table next to the armchair
side_table[ ].position = [. ,
. , side_table[ ].size[ ] /]

floor_lamp[ ].position = [. ,
. , floor_lamp[ ].size[ ] /]

Spatial Relations

solver.align_to_center(table, rug[ ])

solver.distance_constraint(table, sofa[ ])

Self-consistency filtering

Optimizer

Stage	What runs	Optimizer mechanism
Main optimization	All movable assets; minimize overlap + constraints + reachability	Adam (gradient descent)
Projection (hard constraints)	After every main iters + at end	Projection
Refinement (cleanup)	Only assets still overlapping after main optimization	Adam, only on problematic assets

Rendering

D Scence Generation